📊 Personal Finance Manager Complete Project Documentation

1️⃣ Project Title

Personal Finance Manager using Python (CLI Based)

2️⃣ Introduction

Managing personal expenses is essential for financial planning. Many individuals fail to track their daily expenses, which leads to overspending and poor money management. The **Personal Finance Manager** is a Python-based command-line application designed to help users record, manage and back up their expenses efficiently. This project demonstrates practical implementation of **Object-Oriented Programming, File Handling, Data Validation, and Reporting**, making it ideal for academic submission and real-world understanding.

3️⃣ Problem Statement

Individuals often do not maintain systematic records of their expenses. Manual tracking is time-consuming and error-prone. There is a need for a simple, reliable, and user-friendly system to:

- Record expenses
- Analyze spending habits
- Generate reports
- Secure data using backups

4️⃣ Objectives of the Project

- To build a CLI-based personal finance management system
- To store expense data persistently using files
- To generate category-wise and monthly expense reports
- To implement search and backup functionality
- To apply modular and structured Python programming

5️⃣ Scope of the Project

This system is suitable for:

- Students and beginners
- Personal daily expense tracking
- Academic mini-projects
- Learning file handling and OOP concepts

The project can be extended to GUI or web-based systems in the future.

6️⃣ Technologies Used

Component	Description
Programming Language	Python 3.8+
Storage	JSON File
Libraries	json, os, datetime, collections
Interface	Command Line Interface

7️⃣ Project Structure

```
finance_manager/
│
├── main.py          # Application entry point
├── expense.py       # Expense class
├── file_manager.py  # File handling & backup
├── reports.py       # Reports and analytics
├── menu.py          # CLI menu handling
├── utils.py         # Input validation utilities
│
├── data/
│   └── expenses.json  # Stored expense records
│
├── backups/         # Backup files
```

```
└── reports/        # Generated reports
```

---

8 Functional Requirements

- Add new expense
- View all expenses
- Category-wise expense summary
- Monthly expense report
- Search expense by category or description
- Backup expense data

---

9 Non-Functional Requirements

- Easy to use
- Fast execution
- Reliable data storage
- Error-free input handling
- Portable across systems

---

10 Modules Description

◆ Add Expense Module

Allows users to enter:

- Amount
- Category
- Description
- Date (auto-filled if empty)

◆ View Expense Module

Displays all expenses in a tabular format sorted by date.

◆ Category-wise Summary Module

Calculates and displays total spending per category.

◆ Monthly Report Module

Generates report for a specific month using YYYY-MM format.

◆ Search Module

Searches expenses using category or description keywords.

◆ **Backup Module**

Creates timestamped backup files to prevent data loss.

1️⃣1️⃣ Sample Data Format

```
{
 "date": "2026-01-20",
 "amount": 500.0,
 "category": "Transport",
 "description": "Bus pass"
}
```

1️⃣2️⃣ Error Handling

- Numeric validation for amount
- Date format validation (YYYY-MM-DD)
- Graceful handling of empty search results
- Input validation for menu selection

1️⃣3️⃣ Testing

- Unit testing by entering valid and invalid inputs
- Boundary testing for date and amount
- Manual testing for menu navigation

1️⃣4️⃣ Advantages

- Simple and user-friendly
- Lightweight and fast
- Secure data backup
- Modular design
- Easy to extend

1️⃣5️⃣ Limitations

- CLI-based interface only
- No multi-user support

- No graphical visualization

---

1 6 Future Enhancements

- GUI using Tkinter or PyQt
- Web-based version using Flask/Django
- Graphical reports using Matplotlib
- Budget alerts and limits
- User authentication

---

1 7 Conclusion

The **Personal Finance Manager** successfully fulfills its objective of helping users track and analyze expenses efficiently. It provides hands-on experience with Python programming concepts and serves as a strong foundation for advanced finance applications.

---

1 8 Author Details

Name: Khushbu Prajapati
Course: IT Engineering
Skills Used: Python, File Handling, OOP